Procurement: the track unit on Nhien's Rev B frame

Date 16 Oct 2026. Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2. The track comes in two options, A and B; everything else is the same whichever is chosen, so section 2 and the pulleys can be ordered first.

1A. Track, option A: my printed track

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl` (drawing TS-01)	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h

Deck: no change needed; the 3 mm cleats sit 5 mm under the deck.

1B. Track, option B: Nhien's bought belt with cleats (Rev B B10)

Part	Qty	Make / buy	Where	Price	Arrives
Track belt HTD 550-5M-15, 110 teeth, 15 wide (e.g. SKF PHG 550-5M-15), drawing TS-01B	4	**Buy**	RS SG, stock no. 058-6535 or MISUMI SG HTD 5M belts	see listing (about S$25-35 each)	RS SG stock: 1-3 days
Cleats, rubber 10 x 7.9 x 15, bonded on the belt back every 5th tooth (25 mm), 22 per belt	88	**Buy + bond**: belt shop	Local belt / conveyor shop (Fenner, Megadyne or Gates distributor), with the belts	quote	allow 1-2 weeks
Deck change for the belt: notch P12 13 x 135 over each front lane (x -25 to 110), or cut the cleats to 5 high	1	**Make**: with the deck laser cut, or file / mill after	Professional Workshop	-	with the deck

1C. Wheels, both options

Part	Qty	Make / buy	Where	Price	Arrives
Timing pulley HTD 5M 20T, 15 mm belt, flanged, **plain (pilot) bore**	8	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x D-bore Ø8 (M1), 4 x Ø16 H7 (M2)	8	**Make**: Professional Workshop request, Rev B sheet 21	E3-01-06	free	workshop queue, allow 1-2 weeks

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or

			a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10)	32	Buy	RS SG stock no. 427-991
Hinge shafts P20 A 82.3 + B 77.3 (or one 212), Ø8 D-shaft flat 7	2	Buy stock, cut	RS SG linear shafts or MISUMI SG
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets P18 (25x25x4 angle) x4, P19 (20x20x3) x4, T-nuts, E-clips, shims	-	Buy / make	Rev B BOM sections 3-4
Deck P12 (slot H 3 mm longer, see pack page 2), tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM

3. Choosing between A and B

Option A is in-house: four 3 h prints, no supplier, 5 mm deck clearance, and the cleat spacing is matched to the ES1 step. Option B is bought: belts from RS in days, but the 88 cleats need a belt shop (1-2 weeks) and the deck needs a notch or shorter cleats. Wheels, bore modification, plates, shafts, bearings and shoulder screws are the same in both.

4. Order of events

1. Order the 8 plain-bore pulleys today (critical path, then the workshop bore mod); for option B also order the 4 belts and send them with TS-01B to the belt shop. For option A print the stand-in pulley and fit-test slice, then the 4 tracks as soon as a UP300 slot is free.
2. Nhien: laser plates and deck (slot H 74 long; notched if option B), order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit the tracks on the real pulleys; tension at the slots.